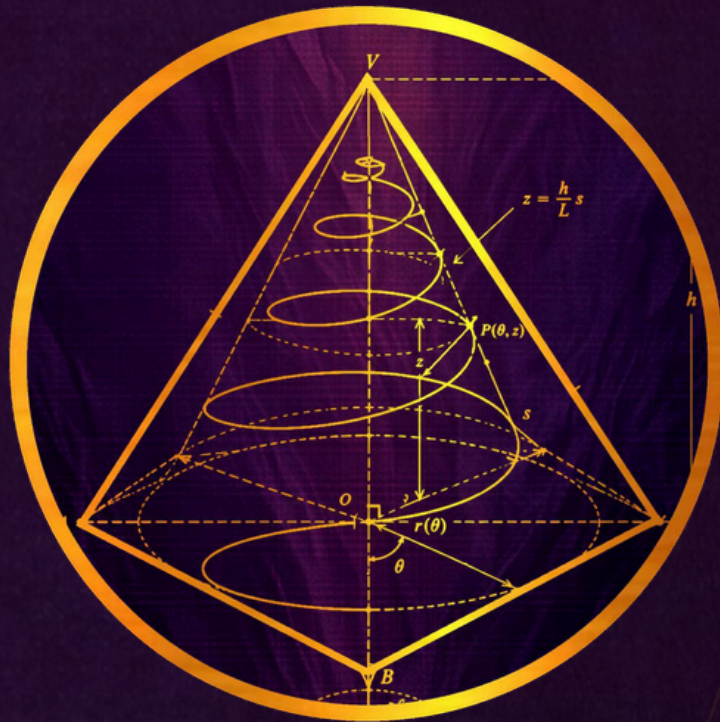




AXIOM OF CHOICE

STUDY GUIDE

INGENIUM

INTRODUCTION

Welcome to the Category guide for Axiom of Choice, a manual for those who are thrilled to test speed, rigour and understanding of core Mathematical fundamentals.

ROUND 1

Following the fair draw, delegates will directly jump into the battle field where two teams face off in a fast paced PVP match. With strategies it's all about staying light on your feet, spotting quick patterns and trusting your problem - solving instincts.

Team Cap: 3 Delegates

The Format: Three simultaneous 1 v 1 duels. The team with the higher score combined from all three duels will eventually move forward.

● DELEGATE CHECKLIST:

- Bring your own mobile devices or laptops with a stable mobile data connection.
- Make sure every team member has an active **Matiks** account ready to go.
- Considering the matches will take place simultaneously, **each team needs at least 3 seperate devices** logged in and ready.

● Skills to Practice:

- Quick mental estimation and pattern spotting.
- Thinking on your feet when clock is ticking.
- Intuitive puzzle navigation over heavy calculations.



ROUND 2

Time to put your strategic minds together! Round 2 is a 90-minute puzzle-solving climb across 5 distinct ranks. Imagine it as navigating an interactive obstacle course of brain teasers, logical hurdles and lateral thinking problems. To make the battleground heated, mystery bombs are dropped every 30 minutes. The top 10 teams who conquer the most ground move on to the finals!

Team Cap: 1-3 Delegates

The Format: Neatly circled questions will be checked, no marks for the solution but in case any team is found cheating, they can be double checked with their solution to maintain fairness.

● DELEGATE CHECKLIST:

○ Carry your own calculator with you, you will be provided with pen and paper (it's preferable to have stationery with you as well)

● Skills to Practice :

- Spotting hidden geometrics, symmetry, and grid-based patterns without needing formulas.
- Untangling tricky word puzzles, sequence riddles, and constraint games using pure reasoning.
- Figuring out how moving parts, counters, or sequences behave when rules shift.

ROUND 3

The Endstufe!! Now teams working is translated from solving puzzled maths to explaining core concepts of how the fundamentals work. Your team will pick an intriguing statement, dive into its core logic, and pitch a creative, well-researched defense to the jury. Whether you use slides, a quick video, or draw it out live on a board, this round is all about storytelling through logic and clear visual reasoning.

Team Cap: 1-2 Delegates

The Format: A live explanation of claims, proving each word said using mathematical ideas and logic in mere 5 minutes.

The Bonus Challenge: Teams scoring over 65% on their main defense unlock an extra point opportunity. Teams will be given 3 minutes to determine a statement and prove your reasoning without any preparation time.

● DELEGATE CHECKLIST:

- Bring with you your presentation or video working smoothly on at least 2 devices to avoid inconvenience. Make sure to carry a tab if you want to present live on it, else you will be provided with a board and marker (You need to inform the median beforehand)
- Carry with you printed research if you want to present it to jury external of the 5 minutes presentation, neat A4 size sheets are recommended to maintain decorum for maths.

● Skills to Practice :

- Breaking down big ideas into simple, beautiful explanations.
- Drawing clear diagrams and visual proofs that instantly make sense to an audience.
- Collaborating smoothly under pressure to defend your team's reasoning.

